Lab 7: Fuzzing for Input Validation Bugs

Student: Dhruv Talavat | **SRN:** PES1UG24AM087

Repository: github.com/DhruvTalavaT27/L6_L7_24_pizza_order (*Lab 7*)

Application under test: pizza_order.py | **Fuzz test file:** test_pizza_order.py | **Library:** Hypothesis (property-based fuzzing)

This document contains the terminal screenshots requested in Steps 2, 4, 5 and 6 of the handout.

Step 2 - run the buggy application (valid input, then malformed input)

Step 4 - run test_pizza_order.py on the buggy application (failing)

Step 5 - run the fixed application (valid input, then the same malformed input)

Step 6 - re-run test_pizza_order.py on the fixed application (passing)

STEP 2 - Run the buggy application (pizza_order.py)

The original script works for valid input, but any unexpected value reaches the dictionary lookups and crashes it.

```
dhruv@pes:~/lab7_orig$ printf 'cheese,olives
M
900/3
' | python pizza_order.py
==== Running toppings_price ====
Enter toppings as CSV (e.g. cheese,olives): Toppings Price: 70

==== Running pizza_inches ====
Enter pizza size (S/M/L): Size in Inches: 10

==== Running split_bill ====
Enter total and people (e.g. 900/3): Per Person: 300.0
dhruv@pes:~/lab7_orig$ printf 'chicken
XL
900/0
' | python pizza_order.py
==== Running toppings_price ====
Enter toppings as CSV (e.g. cheese,olives): Traceback (most recent call last):
  File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 34, in <module>
    main()
    ~~~~^^
  File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 20, in main
    result = toppings_price(user_input)
  File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 7, in toppings_price
    total += TOPPINGS[t.strip().lower()]
             ~~~~~~~~~^^^^^^^^^^^^^^^^^^^
KeyError: 'chicken'
```

STEP 4 - Run test_pizza_order.py on the buggy application (tests fail)

Hypothesis generated the malformed inputs automatically; 21 tests fail against the unfixed script.

```
dhruv@pes:~/lab7_orig$ python -m pytest test_pizza_order.py -q --tb=no
FFFFFFFFFFFFFFFFFFFF.......                                              [100%]
========================== short test summary info ==========================
FAILED test_pizza_order.py::test_toppings_price_no_crash - ExceptionGroup: Hy...
FAILED test_pizza_order.py::test_pizza_inches_no_crash - ExceptionGroup: Hypo...
FAILED test_pizza_order.py::test_split_bill_no_crash - ExceptionGroup: Hypoth...
FAILED test_pizza_order.py::test_toppings_price_rejects_junk_without_crashing[None]
FAILED test_pizza_order.py::test_toppings_price_rejects_junk_without_crashing[]
FAILED test_pizza_order.py::test_toppings_price_rejects_junk_without_crashing[   ]
FAILED test_pizza_order.py::test_toppings_price_rejects_junk_without_crashing[,]
FAILED test_pizza_order.py::test_toppings_price_rejects_junk_without_crashing[,,]
FAILED test_pizza_order.py::test_toppings_price_rejects_junk_without_crashing[chicken]
FAILED test_pizza_order.py::test_pizza_inches_unknown_size_returns_zero[None]
FAILED test_pizza_order.py::test_pizza_inches_unknown_size_returns_zero[] - K...
FAILED test_pizza_order.py::test_pizza_inches_unknown_size_returns_zero[XL]
FAILED test_pizza_order.py::test_pizza_inches_unknown_size_returns_zero[small]
FAILED test_pizza_order.py::test_pizza_inches_unknown_size_returns_zero[12]
FAILED test_pizza_order.py::test_split_bill_unparseable_input_returns_zero[None]
FAILED test_pizza_order.py::test_split_bill_unparseable_input_returns_zero[]
FAILED test_pizza_order.py::test_split_bill_unparseable_input_returns_zero[abc]
FAILED test_pizza_order.py::test_split_bill_unparseable_input_returns_zero[900]
FAILED test_pizza_order.py::test_split_bill_unparseable_input_returns_zero[900/0]
FAILED test_pizza_order.py::test_split_bill_unparseable_input_returns_zero[900/0/3]
FAILED test_pizza_order.py::test_split_bill_unparseable_input_returns_zero[-900/3]
21 failed, 7 passed in 5.18s
dhruv@pes:~/lab7_orig$ python -m pytest test_pizza_order.py -q -k no_crash --tb=line
FFF                                                                      [100%]
================================== FAILURES ===================================
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 26, in test_toppings_price_no_crash
  |     def test_toppings_price_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 28, in test_toppings_price_no_crash
    |     result = toppings_price(s)
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 7, in toppings_price
    |     total += TOPPINGS[t.strip().lower()]
    |              ~~~~~~~~^^^^^^^^^^^^^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 30, in test_toppings_price_no_crash
    |     assert False, f"toppings_price crashed with: {e!r}"
    | AssertionError: toppings_price crashed with: KeyError('')
    | assert False
    | Failing test case: test_toppings_price_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 28, in test_toppings_price_no_crash
    |     result = toppings_price(s)
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 6, in toppings_price
    |     for t in toppings_csv.split(','):
    |              ^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 30, in test_toppings_price_no_crash
    |     assert False, f"toppings_price crashed with: {e!r}"
    | AssertionError: toppings_price crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_toppings_price_no_crash(
    |     s=None,
    | )
```

```
    +------------------------------------
C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py:26: ExceptionGroup: Hypothesis found 2 distinct failures. (2
sub-exceptions)
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 36, in test_pizza_inches_no_crash
  |     def test_pizza_inches_no_crash(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 38, in test_pizza_inches_no_crash
    |     result = pizza_inches(s)
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 11, in pizza_inches
    |     return SIZES[size.strip().upper()]
    |            ~~~~~^^^^^^^^^^^^^^^^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 40, in test_pizza_inches_no_crash
    |     assert False, f"pizza_inches crashed with: {e!r}"
    | AssertionError: pizza_inches crashed with: KeyError('')
    | assert False
    | Failing test case: test_pizza_inches_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 38, in test_pizza_inches_no_crash
    |     result = pizza_inches(s)
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 11, in pizza_inches
    |     return SIZES[size.strip().upper()]
    |                  ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 40, in test_pizza_inches_no_crash
    |     assert False, f"pizza_inches crashed with: {e!r}"
    | AssertionError: pizza_inches crashed with: AttributeError("'NoneType' object has no attribute 'strip'")
    | assert False
    | Failing test case: test_pizza_inches_no_crash(
    |     s=None,
    | )
    +------------------------------------
C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py:36: ExceptionGroup: Hypothesis found 2 distinct failures. (2
sub-exceptions)
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 46, in test_split_bill_no_crash
  |     def test_split_bill_no_crash(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 48, in test_split_bill_no_crash
    |     result = split_bill(s)
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 15, in split_bill
    |     return round(int(total) / int(people), 2)
    |                  ~~~^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 50, in test_split_bill_no_crash
    |     assert False, f"split_bill crashed with: {e!r}"
    | AssertionError: split_bill crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_split_bill_no_crash(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
```

```
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 48, in test_split_bill_no_crash
    |     result = split_bill(s)
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 14, in split_bill
    |     total, people = text.split('/')
    |     ^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 50, in test_split_bill_no_crash
    |     assert False, f"split_bill crashed with: {e!r}"
    | AssertionError: split_bill crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_split_bill_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 48, in test_split_bill_no_crash
    |     result = split_bill(s)
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\pizza_order.py", line 14, in split_bill
    |     total, people = text.split('/')
    |                     ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py", line 50, in test_split_bill_no_crash
    |     assert False, f"split_bill crashed with: {e!r}"
    | AssertionError: split_bill crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_split_bill_no_crash(
    |     s=None,
    | )
    +-----------------------------------
C:\Users\Dhruv\AppData\Local\Temp\lab7_orig\test_pizza_order.py:46: ExceptionGroup: Hypothesis found 3 distinct failures. (3
sub-exceptions)
========================== short test summary info ==========================
FAILED test_pizza_order.py::test_toppings_price_no_crash - ExceptionGroup: Hy...
FAILED test_pizza_order.py::test_pizza_inches_no_crash - ExceptionGroup: Hypo...
FAILED test_pizza_order.py::test_split_bill_no_crash - ExceptionGroup: Hypoth...
3 failed, 25 deselected in 0.70s
```

STEP 5 - Run the fixed application (pizza_order.py)

Every function now validates its input, so the same malformed values are handled instead of crashing.

```
dhruv@pes:~/L6_L7_24_pizza_order/Lab 7$ printf 'cheese,olives
M
900/3
' | python pizza_order.py
==== Running toppings_price ====
Enter toppings as CSV (e.g. cheese,olives): Toppings Price: 70

==== Running pizza_inches ====
Enter pizza size (S/M/L): Size in Inches: 10

==== Running split_bill ====
Enter total and people (e.g. 900/3): Per Person: 300.0
dhruv@pes:~/L6_L7_24_pizza_order/Lab 7$ printf 'chicken
XL
900/0
' | python pizza_order.py
==== Running toppings_price ====
Enter toppings as CSV (e.g. cheese,olives): Toppings Price: 0

==== Running pizza_inches ====
Enter pizza size (S/M/L): Size in Inches: 0

==== Running split_bill ====
Enter total and people (e.g. 900/3): Per Person: 0.0
```

STEP 6 - Re-run test_pizza_order.py on the fixed application (all tests pass)

```
dhruv@pes:~/L6_L7_24_pizza_order/Lab 7$ python -m pytest test_pizza_order.py
............................                                             [100%]
28 passed in 2.40s
```

Files submitted with this document

```
Lab 7/pizza_order.py          Fixed application script (all input validation bugs repaired)
Lab 7/test_pizza_order.py     Updated Hypothesis fuzz test file (3 fuzz properties + 25 regression/value tests, 28 tests)
Lab 7/Lab_7_FuzzTesting_PES1UG24AM087.pdf   This PDF with the Step 2, 4, 5 and 6 screenshots
```

Bugs found by fuzzing and fixed: toppings_price (None -> AttributeError, unknown/blank topping -> KeyError), pizza_inches (None -> AttributeError, unknown size -> KeyError), split_bill (None -> AttributeError, no '/' -> ValueError unpacking, non-numeric or empty field -> ValueError, zero/negative divisor -> ZeroDivisionError).